

Celestial Holography and Calabi–Yau Quantum Groups

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Abstract

We develop the connection between celestial holography and the quantum vertex chiral groups of Calabi–Yau threefolds. The Mellin transform of 4d scattering amplitudes produces a 2d celestial chiral algebra on S^2 whose symmetry structure is governed by an infinite-dimensional algebra. For 4d $\mathcal{N} = 2$ theories arising from CY3 compactification, this celestial chiral algebra is identified with the genus-0 sector of the quantum vertex chiral group $G(X)$. The simplest case — the $\mathfrak{w}_{1+\infty}$ symmetry of pure gravity (Strominger et al.) — is recovered as $\mathcal{W}_{1+\infty} = Y(\widehat{\mathfrak{gl}}_1) = G(\mathbb{C}^3)$, the quantum vertex chiral group of flat space. The celestial OPE is the collision r -matrix $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_A)$, and higher soft theorems correspond to higher-arity shadows in the Postnikov tower of the universal Maurer–Cartan element Θ_A . Higher-genus corrections in the shadow obstruction tower control loop-level celestial amplitudes.

Contents

1 Celestial amplitudes and the celestial chiral algebra

1.1 The Mellin transform to the celestial sphere

Four-dimensional scattering amplitudes in Minkowski space $\mathbb{R}^{1,3}$ can be reformulated as correlation functions on the celestial sphere $S^2 \cong \mathbb{P}^1$ at null infinity. The passage is via the Mellin transform in the energy variable.

A massless particle of momentum p^μ can be parametrized as

$$p^\mu = \varepsilon \omega q^\mu(z, \bar{z}), \quad q^\mu(z, \bar{z}) = (1 + z\bar{z}, z + \bar{z}, -i(z - \bar{z}), 1 - z\bar{z}), \quad (1)$$

where $\omega > 0$ is the energy, $\varepsilon = \pm 1$ labels incoming/outgoing, and $(z, \bar{z}) \in \mathbb{P}^1$ is the point on the celestial sphere determined by the direction of propagation. An n -particle scattering amplitude $\mathcal{A}_n(\omega_i, z_i, \bar{z}_i)$ is Mellin-transformed to a *celestial amplitude*

$$\tilde{\mathcal{A}}_n(\Delta_i, z_i, \bar{z}_i) = \prod_{i=1}^n \int_0^\infty d\omega_i \omega_i^{\Delta_i-1} \mathcal{A}_n(\omega_i, z_i, \bar{z}_i), \quad (2)$$

where $\Delta_i \in \mathbb{C}$ is the *conformal dimension* of the i -th operator on the celestial sphere. The Lorentz group $\text{SL}(2, \mathbb{C})$ acts on $\mathbb{P}^1$ by Möbius transformations, and $\tilde{\mathcal{A}}_n$ transforms as a correlation function of primary operators of weights $(\Delta_i, \bar{\Delta}_i)$ in a 2d CFT on $\mathbb{P}^1$.

Definition 1.1 (Celestial chiral algebra). The *celestial chiral algebra* $\mathcal{A}_{\text{cel}}$ of a 4d massless theory is the holomorphic sector of the 2d CFT on $\mathbb{P}^1$ whose correlation functions are the celestial amplitudes (??). This is a vertex algebra on $\mathbb{P}^1$ whose OPE encodes the collinear singularities of the 4d S-matrix.

Remark 1.2 (Collinear limits and the celestial OPE). In the momentum-space S-matrix, collinear singularities arise when two momenta become parallel: $p_i \parallel p_j$. On the celestial sphere, this corresponds to $z_i \rightarrow z_j$, which is exactly the OPE limit. The celestial OPE therefore encodes the universal collinear behavior of the S-matrix:

$$\mathcal{O}_{\Delta_1}(z_1) \cdot \mathcal{O}_{\Delta_2}(z_2) \sim \sum_k \frac{C_k(\Delta_1, \Delta_2)}{(z_1 - z_2)^{h_k}} \mathcal{O}_{\Delta_k}(z_2),$$

where the OPE coefficients C_k are determined by the collinear splitting functions of the 4d theory.

1.2 Soft theorems as Ward identities

The deep structural insight of celestial holography (Strominger 2014, He–Lysov–Mitra–Strominger 2015) is that *soft theorems are Ward identities* of the celestial chiral algebra.

- **Weinberg’s leading soft graviton theorem** (1965): In the limit where one graviton’s energy $\omega \rightarrow 0$, the amplitude factorizes universally. On the celestial sphere, this becomes the Ward identity of a spin-2 current, which generates the supertranslation subalgebra of the BMS group.
- **Cachazo–Strominger subleading soft theorem** (2014): The $\mathcal{O}(\omega^0)$ correction in the soft limit gives a subleading Ward identity, generating the superrotation (Virasoro) subalgebra of the extended BMS group.
- **Higher soft theorems**: Each sub k -leading soft theorem at $\mathcal{O}(\omega^{k-1})$ produces a Ward identity from a spin- $(k+1)$ current in the celestial chiral algebra.

Proposition 1.3 (Soft theorems and the arity filtration). *The soft graviton theorems are in bijection with the arity filtration of the shadow obstruction tower of Vol I. The sub k -leading soft theorem is the Ward identity of the arity- $(k+2)$ shadow $\Theta_{\mathcal{A}}^{(k+2)}$, evaluated on genus-0 conformal blocks via the shadow representation:*

$$\text{Sh}_{0,n}(\Theta_{\mathcal{A}}^{(k+2)}) \cdot \Phi = 0 \quad (\text{arity-}(k+2) \text{ soft Ward identity}).$$

This follows from the arity decomposition of the modular shadow connection (Vol I, Theorem ??). The leading soft theorem is $\text{Sh}_{0,n}(\kappa(\mathcal{A}))$, the subleading is $\text{Sh}_{0,n}(\mathfrak{C}(\mathcal{A}))$, the sub-subleading is $\text{Sh}_{0,n}(\mathfrak{Q}(\mathcal{A}))$, etc.

2 The $\mathfrak{w}_{1+\infty}$ symmetry and $G(\mathbb{C}^3)$

2.1 Strominger’s $\mathfrak{w}_{1+\infty}$

Guevara, Himwich, Pate, and Strominger (2021) established that the leading celestial symmetry algebra for Einstein gravity in 4d is $\mathfrak{w}_{1+\infty}$, the area-preserving diffeomorphisms of the

celestial sphere — the classical (large- c) limit of the $\mathcal{W}_{1+\infty}$ algebra. At the quantum level, the celestial symmetry algebra of gravity coupled to matter at central charge c is the full $\mathcal{W}_{1+\infty}$ algebra.

The $\mathcal{W}_{1+\infty}$ algebra at central charge c is generated by a tower of higher-spin currents $W^{(s)}(z)$ for $s = 1, 2, 3, \dots$, with:

- $W^{(1)}(z)$: a U(1) current (spin 1),
- $W^{(2)}(z) = T(z)$: the stress tensor (spin 2),
- $W^{(s)}(z)$: spin- s currents for all $s \geq 3$.

The OPE among these currents is determined by a single parameter c (the central charge) up to isomorphism, and the algebra is freely generated in the sense that there are no null relations among the generators at generic c .

2.2 The identification $\mathcal{W}_{1+\infty} = Y(\widehat{\mathfrak{gl}}_1) = G(\mathbb{C}^3)$

Three seemingly independent constructions converge on the same algebraic object:

Theorem 2.1 (The triple identification). *There are canonical isomorphisms*

$$\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1) \simeq G(\mathbb{C}^3), \quad (3)$$

where:

- (i) $\mathcal{W}_{1+\infty}$ is the unique one-parameter family of vertex algebras freely generated by one current in each spin $s \geq 1$;
- (ii) $Y(\widehat{\mathfrak{gl}}_1)$ is the affine Yangian of $\mathfrak{gl}_1$;
- (iii) $G(\mathbb{C}^3)$ is the quantum vertex chiral group of the toric CY3 $\mathbb{C}^3$ (Definition ??).

Proof sketch. The isomorphism $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is the theorem of Maulik–Okounkov (2019) and Schiffmann–Vasserot (2013), established through the action of the affine Yangian on the equivariant cohomology of the Hilbert scheme of points on $\mathbb{C}^2$.

The isomorphism $Y(\widehat{\mathfrak{gl}}_1) \simeq G(\mathbb{C}^3)$ follows from the Schiffmann–Vasserot theorem (Vol III, Theorem ??): the critical CoHA of $\mathbb{C}^3$ (Jordan quiver, cubic potential $W = \text{tr}(XYZ - XZY)$) is isomorphic to the positive half $Y^+(\widehat{\mathfrak{gl}}_1)$, and the quantum vertex chiral group $G(\mathbb{C}^3)$ is by definition the full affine Yangian associated to the root datum of $\mathbb{C}^3$. $\square$

Remark 2.2 (Physical content). The identification (??) has the following physical reading:

- (a) The celestial symmetry algebra of 4d gravity ($\mathcal{W}_{1+\infty}$) is the quantum vertex chiral group of flat space ($G(\mathbb{C}^3)$).
- (b) The infinite tower of higher-spin celestial symmetries ($W^{(s)}$ for $s = 1, 2, 3, \dots$) is the tower of generators of the affine Yangian, which in turn is the tower of BPS states in the DT theory of $\mathbb{C}^3$.

- (c) The celestial central charge c is *not* the modular characteristic $\kappa(G(\mathbb{C}^3))$ in general: κ depends on the subalgebra (Heisenberg: $\kappa = c$; Virasoro: $\kappa = c/2$; full $\mathcal{W}_{1+\infty}$: divergent). The DT genus-1 amplitude is controlled by the Heisenberg $\kappa = c$; see the anomaly cancellation note for the distinction.

The simplest celestial symmetry algebra IS the quantum vertex chiral group of the simplest CY3.

3 CY3 compactification and celestial chiral algebras

3.1 4d $\mathcal{N} = 2$ theories from CY3 compactification

Type IIB string theory compactified on a Calabi–Yau threefold X produces a 4d $\mathcal{N} = 2$ supergravity theory whose low-energy effective action is determined by the geometry of X . The vector multiplet moduli space is the complex structure moduli space $\mathcal{M}_{\text{cs}}(X)$; the hypermultiplet moduli space involves the complexified Kähler moduli together with RR flux data.

The BPS spectrum of the 4d theory is controlled by Donaldson–Thomas invariants:

$$\Omega(\gamma; X) = \text{DT}_\gamma(X), \quad \gamma \in H_3(X, \mathbb{Z}), \quad (4)$$

where $\Omega(\gamma; X)$ is the BPS index (second helicity supertrace) of charge γ , and $\text{DT}_\gamma(X)$ is the generalized DT invariant counting coherent sheaves / special Lagrangians / D-branes of class γ .

Conjecture 3.1 (CY3 celestial chiral algebra). *For a 4d $\mathcal{N} = 2$ theory T_X arising from Type IIB on a CY3 X , the celestial chiral algebra is*

$$\mathcal{A}_{\text{cel}}(T_X) \simeq G(X)^{g=0}, \quad (5)$$

the genus-0 (tree-level) sector of the quantum vertex chiral group $G(X)$, with higher-genus corrections from the shadow obstruction tower.

Definition 3.2 (Quantum vertex chiral group). The *quantum vertex chiral group* $G(X)$ of a CY3 X is the E_2 -chiral algebra whose:

- (i) **Root datum** is the generalized root datum $\mathcal{R}(X) = (\Lambda(X), \Delta^{\text{re}}, \Delta_0^{\text{im}}, \Delta_1^{\text{im}}, W(X), \rho, \text{mult})$ extracted from the CY geometry (lattice from $K_0(\text{Coh}(X))$, root multiplicities from DT invariants);
- (ii) **Positive half** is the critical CoHA: $U(\mathfrak{n}_+) = \text{CoHA}(X) = Y^+(\widehat{\mathfrak{g}}_{Q_X})$ for toric X ;
- (iii) **E_2 -structure** is the braided monoidal structure on $\text{Rep}^{E_2}(G(X))$, with braiding from the quantum group R -matrix of the affine super Yangian;
- (iv) **Automorphic form** is the denominator identity of the associated BKM superalgebra: Φ_X (the DT partition function for toric CY3, the Igusa cusp form Δ_5 for $K3 \times E$).

3.2 The genus filtration and loop corrections

The quantum vertex chiral group $G(X)$ has a natural genus filtration inherited from the shadow obstruction tower of the associated chiral algebra $A_X = \Phi(\text{Coh}(X))$:

$$G(X)^{g=0} \subset G(X)^{g \leq 1} \subset G(X)^{g \leq 2} \subset \dots \subset G(X). \quad (6)$$

- **Genus 0** ($G(X)^{g=0}$): The *tree-level* quantum vertex chiral group. This is the sector controlled by the collision r -matrix and the genus-0 shadow obstruction tower. For toric CY3, it is the Yangian R -matrix sector. This is the celestial chiral algebra proper.
- **Genus 1** ($G(X)^{g \leq 1}$): One-loop corrections. The genus-1 obstruction $\text{obs}_1 = \kappa(A_X) \cdot \lambda_1$ produces the first quantum correction to the celestial OPE. The modular characteristic $\kappa(A_X)$ controls the leading soft graviton factor (not to be confused with the celestial central charge c , which is a distinct quantity; see Remark ??).
- **Genus g** : The genus- g shadow obstruction tower component $\text{obs}_g(A_X) = \kappa(A_X) \cdot \lambda_g$ (on the uniform-weight lane) produces g -loop corrections. The full tower reproduces the all-loop celestial amplitude.

Proposition 3.3 (Genus-0 celestial amplitudes from $G(X)$). *The tree-level celestial n -point amplitude of the $4d$ $\mathcal{N} = 2$ theory T_X is the genus-0, n -point conformal block of the quantum vertex chiral group:*

$$\tilde{\mathcal{A}}_n^{\text{tree}}(\Delta_i, z_i) = \langle \mathcal{O}_{\Delta_1}(z_1) \cdots \mathcal{O}_{\Delta_n}(z_n) \rangle_{G(X)^{g=0}}.$$

The collinear singularity structure is controlled by the collision r -matrix $r(z)$ of $G(X)$.

4 The soft sector and the modular characteristic κ

4.1 Leading soft theorems from κ

In the Vol I framework, the modular characteristic $\kappa(\mathcal{A})$ is the arity-2 projection of the universal MC element $\Theta_{\mathcal{A}}$. In the celestial context, κ controls the leading soft behavior.

Theorem 4.1 (Leading soft theorem = arity-2 projection). *The Weinberg leading soft graviton theorem for the $4d$ theory T_X is the arity-2 Ward identity of the shadow connection:*

$$\text{Sh}_{0,n}(\kappa(A_X)) \cdot \Phi = \sum_{i < j} \frac{\kappa_{ij}}{z_i - z_j} \Phi = 0, \quad (7)$$

where $\kappa_{ij} = \kappa(A_X)(e_i, e_j)$ is the bilinear form on charge vectors determined by the modular characteristic, and Φ is a celestial correlator (horizontal section of $\nabla_{0,n}^{\text{hol}}$).

For $X = \mathbb{C}^3$: $\kappa(G(\mathbb{C}^3)) = \kappa(H_1) = 1$ (the Heisenberg level, not $c/2 = 1/2$; see AP48), recovering the standard graviton soft factor.

4.2 Higher soft theorems from higher-arity shadows

Each arity level $r \geq 3$ of the shadow obstruction tower contributes a higher soft theorem.

Theorem 4.2 (Higher soft theorems = higher shadows). *The sub^{r-2} -leading soft theorem of the 4d theory T_X is the arity- r Ward identity:*

$$\text{Sh}_{0,n}(\Theta_{A_X}^{(r)}) \cdot \Phi = - \sum_{\substack{s+t=r \\ s,t \geq 2}} [\text{Sh}_{0,n}(\Theta^{(s)}), \text{Sh}_{0,n}(\Theta^{(t)})] \cdot \Phi, \quad (8)$$

where the right side involves the self-interaction of lower-arity shadows. The dictionary is:

Arity	Shadow	Soft theorem	Celestial current
$r = 2$	$\kappa(A_X)$	Weinberg leading	Supertranslation
$r = 3$	$\mathfrak{C}(A_X)$	Cachazo–Strominger subleading	Superrotation (Virasoro)
$r = 4$	$\mathfrak{Q}(A_X)$	Sub-subleading	Spin-3 $\mathcal{W}$ -current
r	$\Theta_{A_X}^{(r)}$	Sub^{r-2} -leading	Spin- $(r-1)$ $\mathcal{W}$ -current

For $X = \mathbb{C}^3$ with shadow depth $r_{\max} = \infty$, the full tower of soft theorems generates the $\mathcal{W}_{1+\infty}$ algebra, recovering the Strominger et al. result.

Remark 4.3 (Shadow depth and the celestial spectrum). The shadow depth $r_{\max}(G(X))$ determines the celestial spin content. For CY3 geometries with finite shadow depth (if any exist), only finitely many higher-spin currents appear; the celestial chiral algebra is a truncation of $\mathcal{W}_{1+\infty}$. For generic CY3, we expect $r_{\max} = \infty$ (class **M**), giving the full infinite tower. The physical reason: the BPS spectrum of a generic CY3 is infinite, and each BPS state contributes a celestial operator.

5 The celestial OPE as collision r -matrix

5.1 The collision r -matrix from Θ_A

The collision r -matrix of a quantum chiral algebra A is the arity-2 genus-0 projection of the universal MC element:

$$r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_A) \in A \otimes A \otimes k((z)), \quad (9)$$

the residue of Θ_A along the collision divisor in $\overline{\mathcal{M}}_{0,3}$ (the moduli space with two coalescing points and one output). This is the E_2 -chiral R -matrix of Vol III, Construction ??.

Warning 5.1 (Bar kernel pole shift). The bar complex extracts residues along $d \log(z_i - z_j)$, not along $dz/(z - w)$. The $d \log$ measure absorbs one power of $(z - w)$; hence $r(z)$ has pole orders one less than the OPE (Vol I, AP19). For a Virasoro OPE with poles at z^{-4} , z^{-2} , z^{-1} , the r -matrix has poles at z^{-3} , z^{-1} .

5.2 The celestial OPE

Theorem 5.2 (Celestial OPE = collision r -matrix). *The celestial OPE of the 4d theory T_X is determined by the collision r -matrix of the quantum vertex chiral group $G(X)$:*

$$\mathcal{O}_{\Delta_1}(z_1) \cdot \mathcal{O}_{\Delta_2}(z_2) \sim r_{G(X)}(z_1 - z_2) \cdot (\mathcal{O}_{\Delta_1} \otimes \mathcal{O}_{\Delta_2})(z_2), \quad (10)$$

where $r_{G(X)}(z)$ is defined by (??) with $A = A_X$ the chiral algebra underlying $G(X)$.

The pole structure of the celestial OPE encodes the shadow depth:

- (i) The number of singular terms is $r_{\max}(G(X)) - 1$.
- (ii) The leading pole is determined by $\kappa(A_X)$: $r(z) \sim \kappa(A_X)/z + \dots$.
- (iii) The subleading poles are determined by the higher shadows: the $(z_1 - z_2)^{-(k+1)}$ pole coefficient is $C_k = \text{Res}_{0,2}^{\text{coll}}(\Theta_{A_X}^{(k+2)})$.

Argument. The celestial OPE is the short-distance expansion of two operator insertions on the celestial sphere $\mathbb{P}^1$. In the shadow-connection framework, this is the monodromy of the flat connection $\nabla_{0,n}^{\text{hol}}$ around the diagonal $z_1 = z_2$. By Vol I (Theorem ??), this monodromy is controlled by $\text{Sh}_{0,n}(\Theta_A)$, and the leading singular part is the collision residue $\text{Res}_{0,2}^{\text{coll}}(\Theta_A) = r(z)$. Each arity- r component of Θ_A contributes a pole of order $r - 1$ in the celestial OPE. $\square$

Remark 5.3 (The quantum Yang–Baxter equation). The r -matrix $r_{G(X)}(z)$ satisfies the quantum Yang–Baxter equation:

$$r_{12}(z_1 - z_2) r_{13}(z_1 - z_3) r_{23}(z_2 - z_3) = r_{23}(z_2 - z_3) r_{13}(z_1 - z_3) r_{12}(z_1 - z_2).$$

In the celestial context, this is the *associativity of the celestial OPE*: the OPE of three operators is independent of the order in which the OPE limits are taken. The QYBE is a consequence of the flatness $(\nabla^{\text{hol}})^2 = 0$, which in turn is the MC equation for Θ_A .

6 The full dictionary: CY3 geometry to celestial holography

The passage from CY3 geometry to celestial holography factors through the quantum vertex chiral group as an intermediary:

<i>CY3 geometry</i>	<i>Quantum vertex chiral group</i>	<i>Celestial holography</i>
X (CY threefold)	$G(X)$ (quantum vertex chiral group)	$\mathcal{A}_{\text{cel}}(T_X)$ (celestial chiral algebra)
$\Lambda(X)$ $K_0(\text{Coh}(X))$	= Root lattice of $G(X)$	Charge lattice of celestial operators
$\text{DT}_\gamma(X)$ (DT invariants)	Root multiplicities of $G(X)$	BPS celestial operator spectrum
$\text{CoHA}(X)$ (critical CoHA)	$U(\mathfrak{n}_+) =$ positive half	Holomorphic collinear sector
$r(z)$ $\text{Res}_{0,2}^{\text{coll}}(\Theta_{A_X})$	= E_2 -chiral R -matrix	Celestial OPE
$\kappa(A_X)$ (modular characteristic)	Arity-2 shadow	Leading soft graviton theorem
$\mathfrak{C}(A_X)$ (cubic shadow)	Arity-3 shadow	Subleading soft theorem
$\Theta_{A_X}^{(r)}$ (arity- r shadow)	Shadow Postnikov obstruction tower	Sub $^{r-2}$ -leading soft theorem
$\text{obs}_g(A_X)$ (genus- g obstruction)	Genus- g shadow obstruction tower	g -loop celestial amplitude
Δ_5 / $Z^{\text{DT}}(X)$ (automorphic form)	Denominator identity	Celestial partition function

6.1 The automorphic correction as the shadow obstruction tower

The central identification of the Vol III programme (VISION.md) is:

$$\boxed{\text{Automorphic correction} = \text{Shadow obstruction tower} = \text{Higher soft theorems.}} \quad (11)$$

In detail: the passage from the “naive” algebra $\mathfrak{g}$ (defined by the real root Gram matrix alone) to the full BKM superalgebra $\mathfrak{g}_X$ (with all imaginary roots and their multiplicities) is the automorphic correction. In the Vol I language, this is the passage from the tree-level r -matrix $r(z)$ (arity 2) to the full MC element Θ_A (all arities). In the celestial language, it is the passage from the leading soft theorem to the full tower of soft theorems:

- **Arity 2** (κ): The leading coefficient = the Weyl vector contribution. Celestially: the Weinberg soft factor.
- **Arity 3** ($\mathfrak{C}$): The first imaginary root corrections. Celestially: the Cachazo–Strominger subleading soft theorem.
- **Arity 4** ($\mathfrak{Q}$): Higher corrections. Celestially: the sub-subleading soft theorem with contact coefficient Q^{contact} .

- **Full tower** (Θ_A): The complete automorphic correction, whose denominator identity is the automorphic form. Celestially: the infinite tower of soft theorems whose consistency (MC equation) is the associativity of the celestial OPE.

6.2 The $K3 \times E$ case: Igusa cusp form and celestial holography

For the CY3 family $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ (K3 surface S , elliptic curve E):

Example 6.1 ($K3 \times E$ celestial holography). The quantum vertex chiral group $G(K3 \times E)$ is the generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$ with:

- (a) Three real simple roots $\delta_1, \delta_2, \delta_3$ with Gram matrix $\begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$.
- (b) Imaginary root multiplicities $\text{mult}(\alpha) = f(nm, l)$ where $f(n, l)$ are Fourier coefficients of the weak Jacobi form $\phi_{0,1}$ (the K3 elliptic genus).
- (c) Denominator identity = the Igusa cusp form Δ_5 of weight 5 for $\text{Sp}_4(\mathbb{Z})$.

The celestial interpretation:

- The modular characteristic $\kappa(G(K3 \times E)) = 5$ (the weight of Δ_5).
- The celestial chiral algebra $\mathcal{A}_{\text{cel}}$ is the genus-0 sector of $\mathfrak{g}_{\Delta_5}$.
- The $\text{Sp}_4(\mathbb{Z})$ symmetry of the Siegel modular form IS the E_2 -braiding — the braided monoidal structure on $\text{Rep}^{E_2}(G(K3 \times E))$.
- The shadow obstruction tower encodes the product side of the denominator identity:

$$\frac{1}{64} \Delta_5 = e^{-2\pi i(\rho, z)} \prod_{\alpha \in \Delta_+} (1 - e^{-2\pi i(\alpha, z)})^{\text{mult}(\alpha)}.$$

Each factor in the product corresponds to a shadow obstruction tower component: the real roots ($\text{mult} = 1$) contribute at low arity, and the imaginary roots contribute at higher arities with multiplicities from $\phi_{0,1}$.

6.3 The toric CY3 case: affine super Yangians

Example 6.2 (Toric CY3 celestial holography). For a toric CY3 X without compact 4-cycles:

- (a) The quantum vertex chiral group $G(X) = Y(\widehat{\mathfrak{g}}_{Q_X})$, the affine super Yangian of the toric quiver.
- (b) The celestial chiral algebra is the genus-0 sector: the Yangian R -matrix determines the celestial OPE.
- (c) For $X = \mathbb{C}^3$: $G(\mathbb{C}^3) = \mathcal{W}_{1+\infty}$, recovering the Strominger et al. result directly.
- (d) For the resolved conifold: the toric diagram has two vertices, the quiver is Klebanov–Witten (two nodes, bidirectional arrows), and $G(X)$ is the affine super Yangian of the corresponding quiver Lie superalgebra. The celestial chiral algebra has two families of higher-spin currents, one for each node.

7 Structural theorems

7.1 The celestial shadow connection

The shadow connection of Vol I, specialized to genus 0 and evaluated on the quantum vertex chiral group $G(X)$, gives the celestial shadow connection:

Definition 7.1 (Celestial shadow connection). The *celestial shadow connection* of the 4d theory T_X is

$$\nabla_n^{\text{cel}} := \nabla_{0,n}^{\text{hol}}|_{A=A_X} = d - \text{Sh}_{0,n}(\Theta_{A_X}), \quad (12)$$

a flat connection on the space of n -point celestial amplitudes $\mathcal{V}_{0,n}(A_X)$. Its flatness $(\nabla_n^{\text{cel}})^2 = 0$ is the MC equation for Θ_{A_X} .

Theorem 7.2 (Celestial Ward identities from the shadow obstruction tower). *The horizontal sections of ∇_n^{cel} are the celestial correlators. The arity decomposition of ∇_n^{cel} produces the full tower of soft graviton theorems:*

$$\nabla_n^{\text{cel}} = \nabla_n^{(2)} + A_n^{(3)} + A_n^{(4)} + \dots, \quad (13)$$

where $\nabla_n^{(2)}$ is the leading (supertranslation) connection and $A_n^{(r)}$ for $r \geq 3$ are the higher soft perturbations. The arity- r flatness equation $\nabla^{(2)}(A^{(r)}) + \sum_{s+t=r} [A^{(s)}, A^{(t)}] + o_r = 0$ is the sub^{r-2} -leading soft theorem with all nonlinear corrections from lower arities.

7.2 The celestial Koszul dual

Conjecture 7.3 (Celestial Koszul duality). *The Koszul dual $G(X)^\dagger$ of the quantum vertex chiral group controls the celestial amplitudes of the dual 4d theory. In the E_2 -chiral Koszul duality framework (Vol III, Conjecture ??):*

$$\text{Rep}^{E_2}(G(X)) \simeq \text{Rep}^{E_2}(G(X)^\dagger)^{\text{rev}}, \quad (14)$$

where rev denotes the reverse braiding.

For $X = \mathbb{C}^3$: $\mathcal{W}_{1+\infty}$ is self-dual (the Feigin–Frenkel involution acts trivially at the self-dual level), and the celestial chiral algebra of 4d gravity is self-dual.

7.3 The E_2 -structure and braided celestial amplitudes

Proposition 7.4 (The E_2 -braiding is the celestial crossing symmetry). *The E_2 -structure on $G(X)$ (the braided monoidal structure on $\text{Rep}^{E_2}(G(X))$) is the crossing symmetry of celestial amplitudes. The braiding*

$$\sigma_{V,W}: V \otimes W \xrightarrow{\sim} W \otimes V$$

in $\text{Rep}^{E_2}(G(X))$ corresponds to the exchange $z_i \leftrightarrow z_j$ of celestial operator positions, implemented by the monodromy of ∇^{cel} around the exchange path in configuration space.

The R -matrix $R(z) \in \text{End}(V \otimes W)((z))$ satisfying the QYBE is the algebraic encoding of this crossing symmetry. For the Yangian $Y(\widehat{\mathfrak{gl}}_1)$, the R -matrix is the rational R -matrix $R(u) = 1 + P/u$ (where P is the permutation operator), giving the simplest crossing kernel for graviton celestial amplitudes.

8 Loop corrections and the genus tower

8.1 The genus expansion of celestial amplitudes

The shadow obstruction tower of the quantum vertex chiral group produces a genus expansion for celestial amplitudes that corresponds to the loop expansion of the 4d S-matrix:

$$\tilde{\mathcal{A}}_n(\Delta_i, z_i) = \sum_{g=0}^{\infty} \hbar^{2g-2} \tilde{\mathcal{A}}_n^{(g)}(\Delta_i, z_i), \quad (15)$$

where $\tilde{\mathcal{A}}_n^{(g)}$ is the genus- g celestial amplitude, computed from the genus- g shadow obstruction tower of $G(X)$.

Theorem 8.1 (Loop corrections from the shadow obstruction tower). *The genus- g celestial amplitude of the 4d theory T_X is controlled by the genus- g obstruction class:*

$$\tilde{\mathcal{A}}_n^{(g)} = \int_{\overline{\mathcal{M}}_{g,n}} \text{obs}_g(A_X) \cdot \langle \mathcal{O}_{\Delta_1}(z_1) \cdots \mathcal{O}_{\Delta_n}(z_n) \rangle_{g,n}, \quad (16)$$

where $\text{obs}_g(A_X) = \kappa(A_X) \cdot \lambda_g$ on the uniform-weight lane, and the integral is over the moduli space $\overline{\mathcal{M}}_{g,n}$ of stable genus- g curves with n marked points.

The all-genera generating function is:

$$F_{\text{cel}}(\hbar) = \sum_{g=1}^{\infty} \hbar^{2g} F_g(A_X) = \kappa(A_X) \cdot (\hat{A}(i\hbar) - 1), \quad (17)$$

where $\hat{A}$ is the $\hat{A}$ -genus and $F_g(A_X) = \kappa(A_X) \cdot \int_{\overline{\mathcal{M}}_g} \lambda_g$ are the shadow obstruction tower amplitudes of Vol I.

Remark 8.2 (The $\hbar$ convention). The factor $\hbar^{2g-2}$ in (??) uses the Vol I convention where $\hbar^{2g-2}$ weights the genus- g contribution. The Feynman-section convention $\hbar^{g-1}$ is related by $\hbar_{\text{there}} = \hbar_{\text{here}}^2$. The leading term $g = 0$ at $\hbar^{-2}$ is the tree-level celestial amplitude.

8.2 Interpretation: GW invariants and celestial loop amplitudes

The shadow obstruction tower amplitudes $F_g(A_X)$ involve the Hodge class λ_g integrated over $\overline{\mathcal{M}}_g$. For a CY3 X , the Gromov–Witten invariants $\text{GW}_{g,n,\beta}(X)$ (counting holomorphic curves of genus g and class β in X through n constraints) are related to the shadow obstruction tower via the CY modular characteristic (Vol III, Theorem CY-D). The celestial loop amplitudes therefore encode the enumerative geometry of X : higher-loop celestial amplitudes “count curves” in the CY3.

9 Comparison with the celestial holography literature

9.1 Relation to Costello–Paquette

Costello and Paquette (2022) construct the celestial chiral algebra of self-dual gravity as a specific vertex algebra (the “BCOV chiral algebra”), derived from the Kodaira–Spencer theory of gravity on a CY3. In our framework:

- The Kodaira–Spencer / BCOV theory on X is the “closed-string” side of the CY category $\text{Coh}(X)$.
- The BCOV chiral algebra on $\mathbb{P}^1$ is a model for the genus-0 sector of $G(X)$.
- The $\mathcal{W}_{1+\infty}$ symmetry discovered by Costello–Paquette for $X = \mathbb{C}^3$ is recovered as $G(\mathbb{C}^3)^{g=0} = Y(\widehat{\mathfrak{gl}}_1) = \mathcal{W}_{1+\infty}$.
- The higher-genus corrections that are absent in the Costello–Paquette construction (which works at the self-dual / tree level) are supplied by the shadow obstruction tower.

9.2 Relation to Guevara–Himwich–Pate–Strominger

Guevara, Himwich, Pate, and Strominger identified the leading celestial symmetry as $\mathfrak{w}_{1+\infty}$ (the classical limit of $\mathcal{W}_{1+\infty}$) and conjectured its promotion to the full quantum $\mathcal{W}_{1+\infty}$ at finite c . Our framework:

- Provides the algebraic mechanism: the quantum vertex chiral group $G(\mathbb{C}^3) = \mathcal{W}_{1+\infty}$ at all c simultaneously (the affine Yangian is defined for arbitrary c).
- Explains the origin of $\mathcal{W}_{1+\infty}$: it arises as the quantum vertex chiral group of the simplest CY3, not as an *ad hoc* symmetry algebra.
- Generalizes to arbitrary CY3: for each X , there is a celestial chiral algebra $G(X)^{g=0}$ that extends $\mathcal{W}_{1+\infty}$ according to the geometry of X .

9.3 Relation to Bu–Kim–Li–Zhang

Bu, Kim, Li, and Zhang (2024) relate the celestial OPE to the Yangian R -matrix in the context of self-dual Yang–Mills theory. In our framework, this is the statement that the celestial OPE is the collision r -matrix: $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_A)$.

10 Conjectures and open problems

Conjecture 10.1 (CY3 celestial universality). *For every compact CY3 X , there exists a quantum vertex chiral group $G(X)$ whose genus-0 sector is the celestial chiral algebra of the $4d \mathcal{N} = 2$ theory T_X :*

$$\mathcal{A}_{\text{cel}}(T_X) \simeq G(X)^{g=0}.$$

The construction applies to all CY3 categories (not just toric or $K3 \times E$) via the CY-to-chiral functor Φ of Vol III, Theorem CY-A.

Conjecture 10.2 (Wall-crossing = gauge equivalence). *The Kontsevich–Soibelman wall-crossing formula for DT invariants, applied to the root multiplicities of $G(X)$, is a gauge equivalence of MC elements in the modular convolution algebra:*

$$\Theta_{A_X}|_{t_+} \sim_{\text{gauge}} \Theta_{A_X}|_{t_-},$$

where $t_{\pm}$ are stability parameters on opposite sides of a wall of marginal stability. The celestial chiral algebra changes by a gauge transformation, not an isomorphism — the celestial OPE

coefficients jump, but the underlying algebraic structure (the QYBE, the MC equation) is preserved.

Conjecture 10.3 (Celestial shadow depth). *The shadow depth $r_{\max}(G(X))$ of the quantum vertex chiral group of a CY3 X is:*

- (i) $r_{\max} = \infty$ for all smooth compact CY3s (the BPS spectrum is generically infinite);
- (ii) $r_{\max} = \infty$ for all toric CY3s;
- (iii) Finite shadow depth can occur only for degenerate limits (e.g. orbifold singularities where the BPS spectrum truncates).

This implies that the celestial chiral algebra of a generic CY3 compactification has the full $\mathcal{W}_{1+\infty}$ -type structure — an infinite tower of higher-spin symmetries.

Conjecture 10.4 (Eight celestial chiral algebras from $K3 \times E$). *The eight diagonal divisor modular forms of the Igusa cusp form programme correspond to eight distinct quantum vertex chiral groups $G(X_N)$ for $X_N = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, $N = 1, \dots, 8$, with:*

- (i) Denominator identities = the eight dd-modular forms;
- (ii) Root multiplicities from twisted-twined $K3$ elliptic genera;
- (iii) Celestial chiral algebras $G(X_N)^{g=0}$ related by orbifolding of the $\mathbb{Z}/N\mathbb{Z}$ action.

The Koszul duality relations among these eight algebras should manifest as “celestial S -dualities” exchanging weak and strong coupling in the 4d theories.

Conjecture 10.5 (Non-toric CY3 and the BPS root datum). *For a non-toric CY3 X (e.g. the quintic threefold in $\mathbb{P}^4$), the quantum vertex chiral group $G(X)$ exists with root datum $\mathcal{R}(X)$ determined by:*

- (i) Lattice: $\Lambda(X) = K_0(\text{Coh}(X))$ (or $K_0(\text{Fuk}(X))$ on the mirror side);
- (ii) Root multiplicities: DT invariants $\text{DT}_\gamma(X)$ (or GV / Gopakumar–Vafa invariants);
- (iii) The “BPS algebra” of Joyce, Kontsevich–Soibelman, and Meinhardt–Reineke as the positive half;
- (iv) Denominator identity: the DT generating function $Z^{\text{DT}}(X, q)$.

The key challenge is that without the toric combinatorics, the quiver description is not available; the critical CoHA must be replaced by a more general motivic Hall algebra construction.

11 Summary: the CY3 celestial programme

The programme developed in this note connects three circles of ideas through the quantum vertex chiral group:

- (1) **CY3 geometry** (DT invariants, BPS spectra, $K3$ elliptic genera, automorphic forms) provides the root datum $\mathcal{R}(X)$ of the quantum vertex chiral group $G(X)$.

- (2) **The shadow obstruction tower / bar-cobar machine** (Vol I) provides the algebraic engine: the universal MC element Θ_A , its arity filtration (the shadow obstruction tower), and the flat shadow connection ∇^{hol} .
- (3) **Celestial holography** provides the physical interpretation: 4d scattering amplitudes on the celestial sphere, organized by soft theorems that are the Ward identities of the shadow connection.

The central results and identifications are:

- (a) The simplest celestial symmetry algebra, $\mathcal{W}_{1+\infty}$, IS the quantum vertex chiral group of $\mathbb{C}^3$: $\mathcal{W}_{1+\infty} = Y(\widehat{\mathfrak{gl}}_1) = G(\mathbb{C}^3)$.
- (b) For CY3 compactifications, the celestial chiral algebra is the genus-0 sector of $G(X)$, with higher-genus corrections from the shadow obstruction tower controlling loop amplitudes.
- (c) The celestial OPE is the collision r -matrix: $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{A_X})$.
- (d) The leading soft theorem is the arity-2 projection κ . Higher soft theorems are higher-arity shadows.
- (e) The automorphic correction (passage from real roots to the full BKM with imaginary roots) IS the shadow obstruction tower (passage from arity-2 κ to the full Θ_A) IS the tower of soft theorems.
- (f) The E_2 -braiding of $G(X)$ is the crossing symmetry of celestial amplitudes, with the quantum Yang–Baxter equation as the algebraic encoding of OPE associativity.